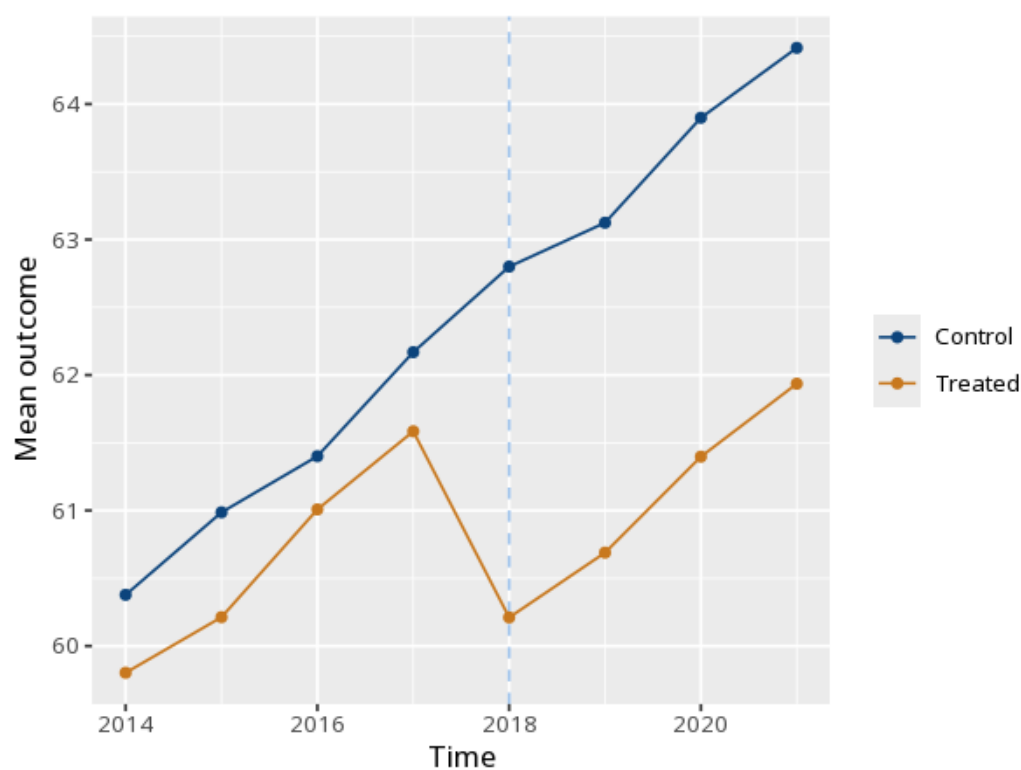


1 Difference-in-differences (DiD)

	(1)
Post	2.33 (0.04) [2.24, 2.41]
Treated $\times$ Post	-1.92 (0.08) [-2.07, -1.76]
Num.Obs.	160
N entities	20
Within R ²	.59
F	F(2, 138) = 98.41, p < .001

Group	Pre	Post
Control	61.23	63.56
Treated	60.65	61.06



The Treated $\times$ Post coefficient is the estimated treatment effect. It rests on the parallel-trends assumption: that the groups would have moved together

absent treatment. Similar pre-treatment trends (inspect the pre-period of the plot) make this more plausible but do not prove it — a visual check is supportive, not confirmatory, since the assumption is about the unobservable post-period counterfactual. The note adds a formal pre-trends test (a pre-period leads-and-lags joint F of the treated $\times$ time interactions): a small p flags diverging pre-trends, while a large p is consistent with parallel trends. Method: R plm package with clustered standard errors (Croissant & Millo, 2008; Bertrand, Duflo & Mullainathan, 2004). Your significance threshold (α) is 0.05.

The DiD estimate was $B = -1.92$, 95% CI $[-2.07, -1.76]$, $p < .001$ (clustered SE).

Statistical basis: Difference-in-differences via entity fixed effects (plm within estimator) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." *Journal of Statistical Software*, 27(2), 1-43.); Clustered standard errors / parallel-trends interpretation (Bertrand, M., Duflo, E., Mullainathan, S. (2004). "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics*, 119(1), 249-275.).